

Kalpana Devi T

Python & AI Developer | Backend Developer | AI Automation Engineer | Generative AI & RAG Developer

Theni, Tamil Nadu, India | +91 82200 93277 | kalpanadevi7e@gmail.com
github.com/kalpanadevi07 | linkedin.com/in/kalpana-devi-4b1140264

PROFESSIONAL SUMMARY

Python and AI Developer with hands-on experience building production AI applications, scalable backend services and automation platforms. Work includes RAG-based chatbots, AI-powered website analysis, Gmail automation, full-stack event platforms, web scraping and computer-vision systems. Focused on transforming complex business requirements into structured, maintainable and practical software solutions.

TECHNICAL SKILLS

Programming & Frontend	Python, JavaScript, C, HTML, CSS, React, Next.js
Backend & APIs	FastAPI, Node.js, Express, REST APIs, OAuth 2.0, token authorization, background tasks
AI & ML	Generative AI, LLM integration, RAG, prompt engineering, LangGraph, NLP, machine learning, computer vision, deep learning
Data & Storage	Supabase, MongoDB, SQL, FAISS, vector databases, data modeling
Libraries & Tools	Pandas, NumPy, Scikit-learn, Keras, OpenCV, Puppeteer, Git, GitHub, Docker, Postman, Swagger UI
Architecture	Modular architecture, scalable system design, configuration-driven development, service-oriented architecture

PROFESSIONAL EXPERIENCE

Python & AI Developer

BrandingBeez, Coimbatore | 2025 - Present

- Built a production AI web-analysis platform that automates website scraping, keyword extraction and SEO report generation using LLM integration, FastAPI and Node.js.
- Deployed a RAG-based AI chatbot using FastAPI, OpenAI API and FAISS with intent classification, brand-tone enforcement and real-time fallback handling.
- Built a Gmail automation system using OAuth 2.0 with automated unsubscribe processing, background-task execution and Swagger UI documentation.
- Developed an AI-powered web scraper and article summarizer that stores structured information in Supabase and displays content through an HTML/CSS frontend.

Data Analytics, Artificial Intelligence and Machine Learning Intern

Novitech R&D; Pvt. Ltd | 2022 - 2024

Completed three focused internships covering data analytics, artificial intelligence and machine learning, gaining experience in model building, feature engineering and data pipelines.

IoT Intern

Iyarkai Tech Lab Pvt. Ltd, Chennai | 2023

Worked on IoT-based embedded systems and sensor-integration projects.

Business Analyst Intern

Sakthi Veera Green Energy Pvt. Ltd, Chennai | 2023

Analysed business processes related to freeze-drying machinery operations and delivered structured reports.

SELECTED PROJECTS

AI-Based Automated Web Analysis System Python · FastAPI · Node.js · LLM Automates website scraping, keyword extraction and structured SEO report generation through modular backend services.	Hybrid AI Chatbot with RAG Architecture Python · FastAPI · OpenAI API · FAISS Production RAG chatbot with intent classification, brand-tone enforcement and fallback handling.
Gmail Automation System FastAPI · Gmail API · OAuth 2.0 Authorized mailbox automation with unsubscribe processing, background execution and Swagger documentation.	AI Event Registration Platform React · Express · MongoDB · Cron Multi-event registration with dynamic forms, live seat countdowns, five-email sequences, reminders and analytics.
AI Web Scraper & Article Summarizer Python · AI/NLP · Supabase Extracts online articles, generates AI-assisted summaries and stores structured results.	IoT Automated Hydroponic System ESP32 · AWS IoT Core · Sensors Sensor-based hydroponic automation concept with cloud-connected real-time monitoring.

COMPUTER VISION LAB

License Plate Recognition OpenCV and Pytesseract OCR with grayscale conversion, Canny edges, contour filtering and ROI extraction.	Face Recognition Real-time OpenCV and Haar Cascade detection with optimized video-frame processing.	Moving Object Detection Frame differencing, background subtraction, contour detection and bounding-box tracking.
--	---	--

EDUCATION

B.E. Computer Science Engineering Jansons Institute of Technology, Coimbatore	2022 - 2026	CGPA: 7.8
Higher Secondary Certificate Scism Matriculation Higher Secondary School	2022	83%
Secondary School Leaving Certificate Scism Matriculation Higher Secondary School	2020	93%

CERTIFICATIONS Programming Languages: C, C++ and Python Data Analytics · Artificial Intelligence · Machine Learning	ACHIEVEMENTS Second Prize - Paper Presentation, Karunya University Top 100 Teams - Hack4Purpose, Anna University, Chennai Technical symposium and inter-college participation
--	---